

Mounika Geriki

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SUMMARY

Applied AI / Machine Learning Engineer with experience building ML models, GenAI/RAG applications, NLP pipelines, and data-driven analytics solutions. Skilled in Python, SQL, PySpark, Scikit-learn, XGBoost, Transformers, LangChain, FAISS, MLflow, Docker, and cloud-based ML workflows. Experienced in fraud/risk analytics, model evaluation, explainability, dashboards, and translating business problems into reliable AI, ML, and analytics solutions.

SKILLS

- Languages:** Python, SQL, PySpark, R, Java, Scala, JavaScript, Bash/Shell, Go, MATLAB, HTML, CSS, C, C++
- Machine Learning:** Scikit-learn, XGBoost, Random Forest, Logistic Regression, Autoencoders, Feature Engineering, Model Evaluation, A/B Testing
- NLP:** PyTorch, TensorFlow, Transformers, BERT, RoBERTa, Hugging Face, NER, Text Classification
- Gen AI / LLMs:** RAG, LangChain, LangGraph, FAISS, Pinecone, Prompt Engineering, LLM Evaluation, RAGAS, LangSmith, OpenAI/Gemini APIs
- Data & Cloud:** Spark, Kafka, AWS S3, SageMaker, GCP BigQuery, Airflow/Composer, Snowflake, PostgreSQL
- MLOps:** MLflow, Docker, FastAPI, Git, GitHub Actions, Model Monitoring, CI/CD
- Analytics:** Pandas, NumPy, Tableau, Power BI, Matplotlib, Excel, Statistical Analysis

PROFESSIONAL EXPERIENCE

- Stony Brook University** | *Graduate Research Assistant* Jan 2025 – Present | Stony Brook, NY
- Built LLM workflows using LangChain to extract structured scientific information from scientific papers and improve literature review efficiency.
 - Evaluated LLM outputs for accuracy, hallucination risk, extraction quality, and consistency across scientific documents.
 - Used prompt engineering, retrieval workflows, and model comparison to improve the reliability of literature-extraction pipelines.
 - Collaborated with research stakeholders to convert unstructured papers into structured datasets for downstream analysis.
 - Worked on preprocessing, document parsing, database organization, and analysis workflows to support applied research use cases.
- Tata Consultancy Services** | *Data Scientist* Sep 2021 – Jan 2024 | Bengaluru, India
- Built and evaluated ML models including Logistic Regression, Random Forest, XGBoost, and Autoencoders to detect fraud patterns and anomalies across large-scale transaction datasets.
 - Performed EDA & feature eng using SQL, Python, and PySpark to identify transaction velocity, frequency, account behavior & fraud-risk indicators.
 - Developed BERT/RoBERTa NLP pipelines for AML/KYC document classification and entity extraction, helping automate compliance reviews.
 - Evaluated models using ROC-AUC, precision-recall, F1-score, confusion matrix, and cross-validation to improve performance on imbalanced data.
 - Applied SHAP and LIME to explain model predictions and communicate fraud-risk drivers to business and compliance stakeholders.
 - Built Tableau and Power BI dashboards to monitor fraud trends, model performance, KPIs, and operational metrics for stakeholder decision-making.
- Infosys** | *Machine Learning Intern* Oct 2020 – Jun 2021 | Bengaluru, India
- Cleaned, transformed, and analyzed structured and unstructured datasets using Python, Pandas, NumPy, and SQL for ML and analytics use cases.
 - Built classification and regression models using Scikit-learn and XGBoost to support forecasting and customer analytics workflows.
 - Developed NLP pipelines using NLTK and spaCy for text classification and sentiment analysis.
 - Created Tableau and Power BI dashboards to track KPIs and communicate insights to business users.

PROJECTS

- InsightPilot – Agentic SQL & Analytics Platform** | *Python, SQL, LangChain, FAISS, Streamlit, OpenAI API*
- Built an agentic AI sys using LangChain to convert business questions into SQL queries, validate outputs, and generate executive-level insights.
 - Added SQL validation guardrails to block unsafe write/delete operations and improve reliability for business users.
- TravelWise – Agentic RAG Travel Assistant** | *Python, Flask, LangChain, FAISS, GCP, REST APIs*
- Built a RAG travel assistant using LangChain, FAISS vector search, and tool-calling for context-aware NYC travel responses.
 - Evaluated the system on 50 queries, achieving 91.4% answer generation, 2.4% error rate, and 2.8s average response time.
- NYC Taxi Data Analytics & Forecasting** | *Python, SQL, BigQuery, GCP Composer, Looker Studio, Scikit-learn*
- Developed scalable ETL pipelines and ML models on GCP, achieving 85% prediction accuracy and improving query performance by 35%

EDUCATION

- Stony Brook University** | Master of Science, Data Science Stony Brook, NY
- Bangalore Institute of Technology** | Bachelor of Engineering, Electronics and Communication Engineering Bengaluru, India

CERTIFICATIONS

- AI Agent Fundamentals (Databricks, Mar 2026) · Google Cloud IAM & Networking for AWS Professionals (Google Cloud, Mar 2026) · Gen AI Intensive Course (Kaggle, Apr 2025) · GenAI in Data Analytics (Meta, Dec 2025)